ADVANCED STATISTICS IN PRACTICE

Implications of Advanced Statistics in Business and Research

Turning data into evidence, decisions, and discovery

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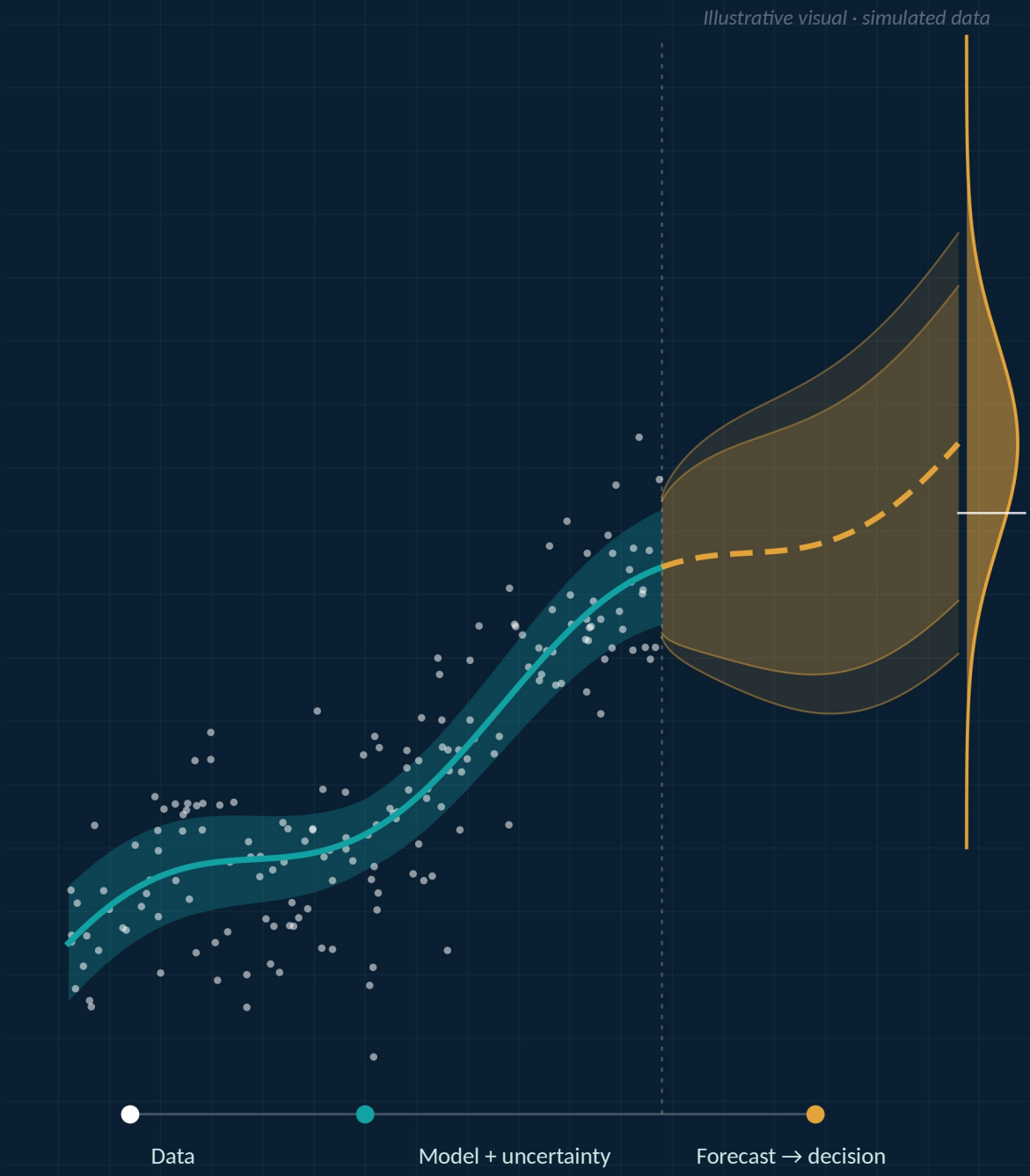
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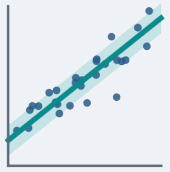


Statistics turns data into decisions

Advanced statistics helps us find real patterns, measure how sure we are, and choose with evidence instead of guesswork.

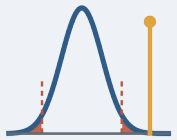
1 Collect data › 2 Find patterns › 3 Measure uncertainty › 4 Make decisions

Four key methods



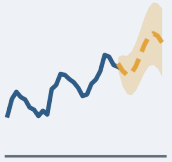
Regression

Shows how one factor affects another, e.g. price → sales.



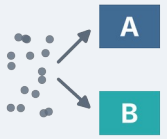
Hypothesis testing

Checks whether a difference is real or just chance.



Forecasting

Predicts future values, such as demand, with a range.



A/B testing

Compares two options fairly using random groups.

Better decisions in business and research

Business

- **Sales forecasting** → plan the right stock level
- **Customer segmentation** → target the right customers
- **Pricing** → find the price that earns most
- **Risk & fraud** → approve loans more safely

REAL CASE

Bing tested longer ad titles with an A/B test: revenue rose about 12% (≈ $100M a year in the U.S.).

Kohavi & Thomke (2017), Harvard Business Review

Research

- **Study design** → make comparisons fair
- **Sample size** → get reliable results
- **Confidence intervals** → show how sure we are
- **Replication** → build results others trust

REAL CASE

RECOVERY trial: dexamethasone cut 28-day deaths in ventilated COVID-19 patients from 41.4% to 29.3%.

RECOVERY Collaborative Group (2021), NEJM

ILLUSTRATIVE DATA hypothetical, not real company results

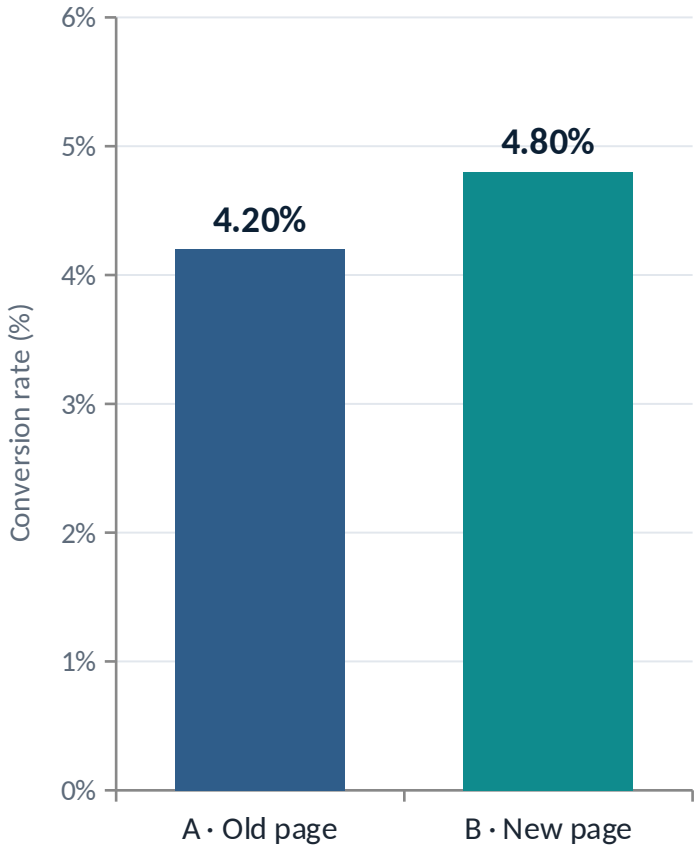
A/B test: does a new product page sell more?

1 Split
40,000 visitors at random: 20,000 see the old page (A), 20,000 the new page (B).

2 Measure
Conversion rate = orders ÷ visitors.

3 Test
Two-proportion z-test at the 5% level.

Visitors are assigned at random, so the only real difference between the groups is the page.



A: 840 / 20,000 · B: 960 / 20,000 orders

+0.60 pp
difference (B − A)

+0.19 to +1.01 pp
95% confidence interval

p = 0.004
very unlikely to be chance

Meaning The new page very likely works better, but the exact gain is uncertain.

DECISION
Launch page B and keep monitoring results.

Use statistics carefully and honestly

WATCH OUT FOR

- **Bad or biased data**
  wrong data gives wrong answers
- **Correlation ≠ causation**
  two things moving together is not proof
- **“Significant” ≠ important**
  always check how big the effect is
- **Privacy & fairness**
  protect people and avoid biased models

KEY TAKEAWAYS

01 **Statistics turns raw data into evidence.**

02 **Testing and measuring uncertainty lead to better decisions.**

03 **Careful interpretation matters as much as the method.**

“The value of data lies not merely in collecting it, but in drawing reliable conclusions from it.”

Thank you · Questions?

References: Kohavi & Thomke (2017), Harvard Business Review 95(5) · RECOVERY Collaborative Group (2021), NEJM 384(8) · Wasserstein & Lazar (2016), The American Statistician 70(2) · Shmueli (2010), Statistical Science 25(3)